Louis Zhang

MECHANICAL ENGINEERING STUDENT - POLYTECHNIQUE MONTREAL

Profile

Mechanical Engineering Student at Polytechnique Montréal with a track record of delivering high-impact prototypes — from lunar transport systems to medical assistive robotics. I bridge the gap between mechanical design and intelligent control to solve real-world engineering challenges through iterative prototyping and technical rigor.

Contact

- louzhang15@gmail.com
- LinkedIn
- Github

Core Compentencies

Domain	Tools & Compentencies
Design	CATIA V6, SolidWorks, GD&T (ASME Y14.5), DFM/DFA, Blender
Analysis	FEA (Ansys, SolidWorks Simulation), MATLAB/Simulink, CFD
Robotics	ROS 2, Python, C++, Sensor Fusion (mmWave, IMU, LIDAR)
Hardware	Arduino, ESP32, Pi Pico, CNC Machining, Rapid Prototyping
Software	Python, C++, React.js, MongoDB, Javascript
Project Management	MS Project, BOM, CSA Compliance Documentation

Portfolio Contents

01 **Lunar Transport System**
CSA LEAP Program · 3 pages

02 **Nursie — AI Nurse Assistant**
ConUHacks X — Best Use of ElevenLabs · 2 pages

03 **Autonomous Reforestation Robot**
RoboHacks 2025 — Unexpected Expedition Award · 1 page

3
Projects

3x
Hackathon wins

Internships

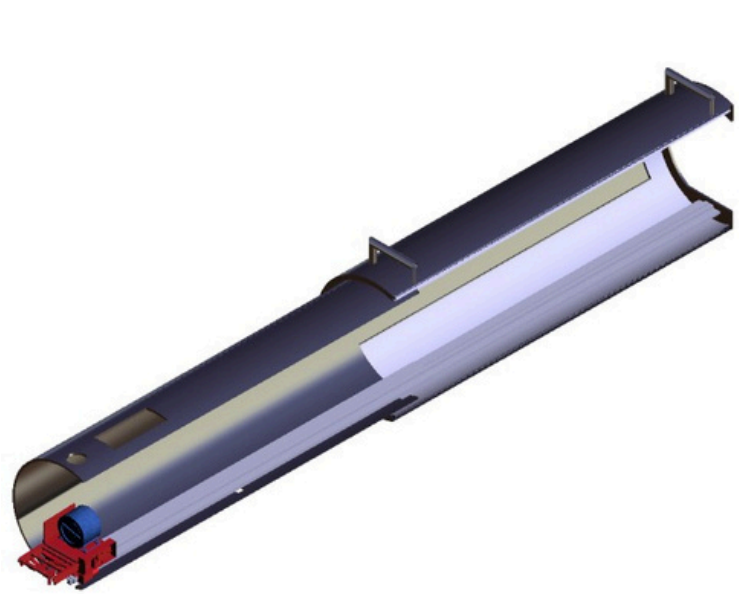
Tesla
Manufacturing Engineering Intern
June 2026 - August 2026

Lockheed Martin
Mechanical Engineering Intern
January 2026 - April 2026

City of Montreal
Engineering Intern
May 2025 - August 2025

CSA LEAP PROJECT PORTFOLIO

LUNAR TRANSPORT SYSTEM



Section Cut CAD — Interior Detail, 3.1m System

- Directed end-to-end conceptualization and holistic integration of a **3.1 m lunar produce-transport system** under the Canadian Space Agency LEAP program. Orchestrated a cross-disciplinary engineering team from concept through physical build to meet **stringent CSA safety profiles.**

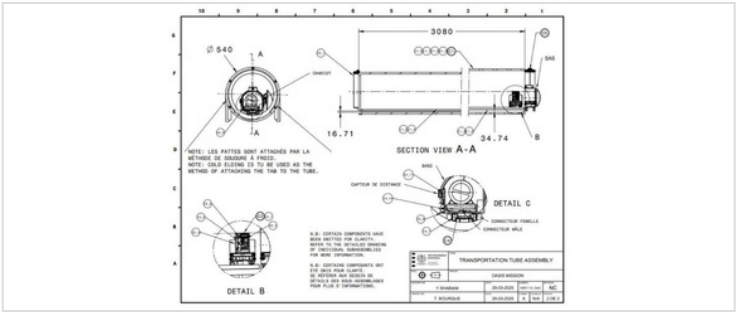
3.1m	−12%	2.5×	50+
Telescopic Envelope	Mass Reduction	Safety Factor	Part Interfaces

PRIMARY TOOLSET

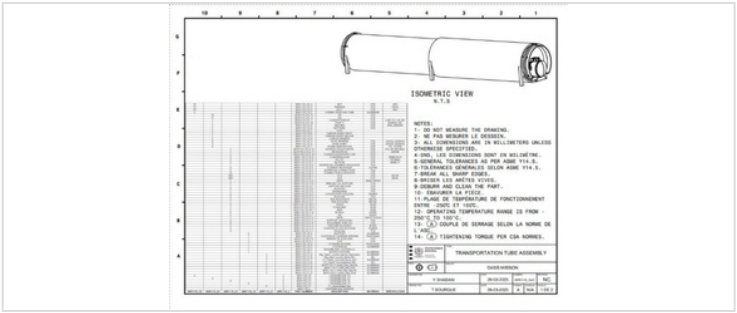
CATIA V6 · SolidWorks · Fusion 360 · GD&T · MS Project · Cold Welding

DESIGN ARCHITECTURE

Constraint Management · Production Specification



TRANSPORTATION TUBE ASSEMBLY — SECTION VIEW A-A



TRANSPORTATION TUBE ASSEMBLY — ISOMETRIC VIEW & BOM

Constraint Management

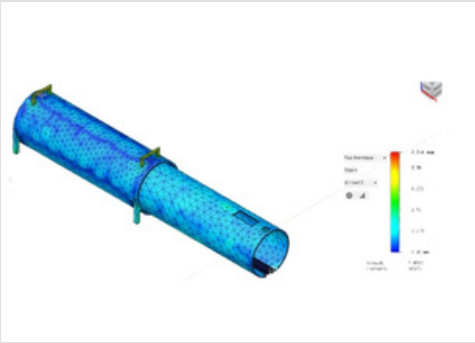
- Managed **50+ unique part interfaces** across the full system envelope — coordinating geometry conflicts between telescopic, structural, and thermal subsystems
- Iteratively optimized part geometry to achieve a **12% system mass reduction** while sustaining a **2.5× structural safety factor** under lunar load conditions
- Used **CATIA V6** parametric assemblies to enforce interface constraints and propagate design changes across dependent geometry
- Coordinated design freeze milestones with a **cross-disciplinary team** using MS Project, tracking interface dependencies across mechanical, thermal, and electrical subsystems

Production Specification

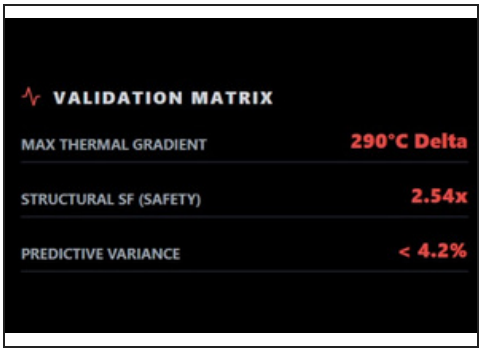
- Applied **GD&T** to all critical mating surfaces — defining flatness, perpendicularity and true-position tolerances for downstream manufacturing
- Defined **cold-welding assembly parameters** for tab-to-tube joints to eliminate thermal fatigue failure points in the vacuum operating environment
- Produced formal **manufacturing drawings** and assembly sequences aligned to CSA build standards
- Conducted tolerance stack-up analysis across all telescopic stages to verify fit under worst-case dimensional variation

FEA · THERMAL · KINEMATIC

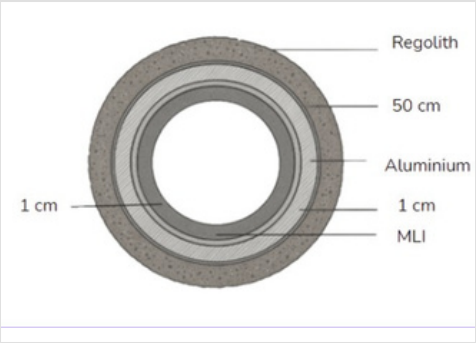
PREDICTIVE SIMULATION & ANALYSIS



THERMAL FEA
Heat distribution — 14-day lunar night cycle simulation



VALIDATION MATRIX
Zero-interference validation across full 3.1m deployment



MATERIAL STACK CROSS-SECTION
High-vibration launch profile — drive mechanism modal

SIMULATION ANALYSES

Thermal Interface Control · Kinematic Integrity

Thermal Interface Control

- Verified **heat retention limits** during 14-day lunar night cycle — ensuring produce viability through the full cold-soak period
- Identified and mitigated **thermal bridging** in telescopic aluminum joints through geometry modifications and MLI placement
- Iterative FEA-guided redesign to isolate conduction paths at each joint interface

Kinematic Integrity

- Validated drive mechanism against **high-vibration launch profiles** — confirmed structural integrity under worst-case launch loads
- Verified **zero mechanical interference** across the full 3.1m telescopic deployment sweep through kinematic motion study
- Confirmed clearance margins at every discrete extension state across the deployment range

MATERIAL STACK-UP ENGINEERING

Outer → Inner

Layer 01 — Regolith Shielding

50cm radiation protection layer utilizing **local lunar regolith** for maximum passive thermal mass. Zero additional launch weight penalty — harvested in-situ.

`THICKNESS: 50cm · SOURCE: In-situ harvest · FUNCTION: Radiation + thermal mass`

Layer 02 — Al 6061 Pressure Vessel

High-strength structural core machined from **Aluminum 6061** — designed for internal atmospheric containment and primary load-bearing under operational pressure.

`MATERIAL: Al 6061-T6 · FUNCTION: Structure + pressure containment · SAFETY: 2.5× factor`

Layer 03 — Multi-Layer Insulation (MLI)

Reflective barriers optimized to minimize **radiative heat transfer** in high-vacuum operating conditions. Positioned at all critical interfaces to break thermal bridging paths.

`TYPE: MLI blanket · FUNCTION: Radiative barrier · ENVIRONMENT: High-vacuum`

EMBEDDED LOGIC · POWER · INTEGRATION

MECHATRONICS & FABRICATION

MECHATRONICS & EMBEDDED LOGIC

Power & Charging Interface

Designed the power routing and charging interface to maintain **electrical continuity across all telescopic extension states** — from stowed to full 3.1m deployment. Validated signal integrity throughout the complete motion envelope.

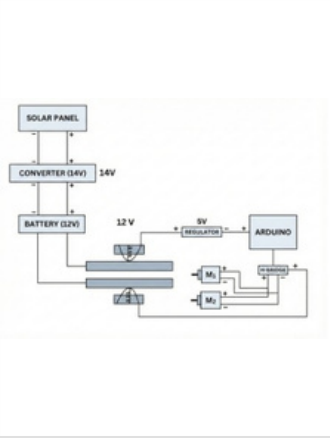
DYNAMIC VALIDATION — CONFIRMED

Electrical continuity verified during live telescopic extension cycle. **Autonomous transport cycle** end-to-end — no manual intervention required.

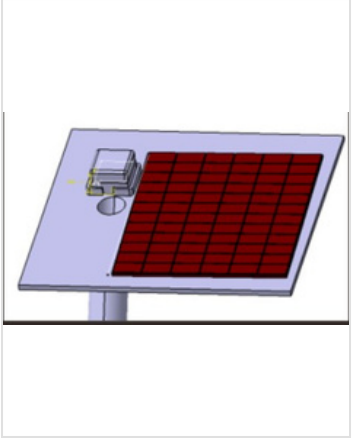
LOGIC STACK INTEGRATION

PRIMARY: ARDUINO (WAGON LOOP)

SECONDARY: PI PICO (CONVEYOR LOGIC)



POWER & CHARGING CIRCUIT



SOLAR PANEL ASSEMBLY

▶ DYNAMIC VALIDATION – VIDEO 1

Rail Conductivity Test

Validates electrical continuity across telescopic rail extension

WATCH VIDEO →

RAIL CONDUCTIVITY TEST

▶ DYNAMIC VALIDATION – VIDEO 2

Full Assembly Demo

End-to-end autonomous transport cycle — full system operation

WATCH VIDEO →

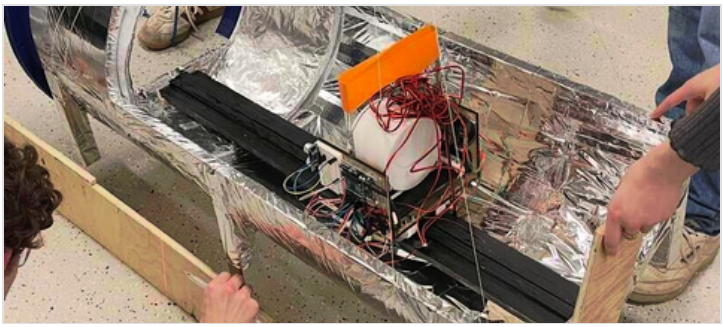
FULL ASSEMBLY DEMO

PHYSICAL FABRICATION & ASSEMBLY

Served as **Lead Physical Integration** — coordinating mechanical, electrical, and thermal assembly sequences across all subsystems to meet CSA build specifications and test readiness milestones.

FABRICATION SPECIALIZATIONS

- AL-6061 Machining
- MLI Fabrication
- PCB Prototyping
- Wiring Harnesses



PHYSICAL INTEGRATION — LAB BUILD

TECHNICAL SUMMARY

At a glance

DISCIPLINE	SCOPE	OUTCOME
Systems Integration	50+ part interfaces, 3.1m envelope	–12% mass, 2.5× safety factor
Thermal Engineering	14-day lunar night cycle FEA	Heat retention verified, bridging mitigated
Mechatronics	PCB, wiring harness, embedded logic	Continuity validated across all states
Fabrication	AL-6061, MLI, PCB, wiring	Lead physical integration — CSA spec

KEY COMPETENCIES DEMONSTRATED

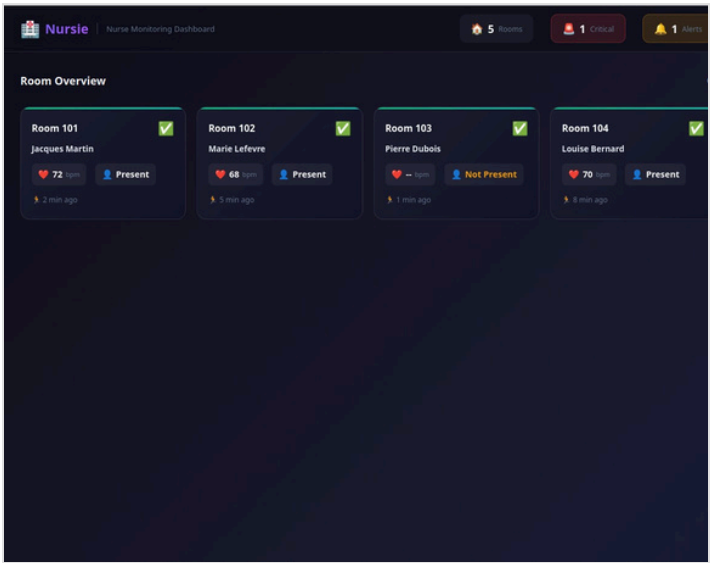
Summary

- ✓ Managed 3.1m telescopic deployment envelope under CSA safety constraints
- ✓ Advanced CAD modelling (CATIA V6) + FEA simulation (thermal & kinematic)
- ✓ Validated electrical continuity across all telescopic motion states
- ✓ Hands-on fabrication lead — machining, PCB, MLI, wiring, full assembly

HACKATHON PROJECT · LONG-TERM CARE · SENSOR FUSION · EDGE AI

BEST USE OF ELEVENLABS (ConUHacks X)

NURSIE
AI NURSE



Nurse Monitoring Dashboard — Real-time room overview & active alerts

Built an **on-premises AI nurse assistant** for Quebec long-term care facilities (CHSLDs) that detects falls and monitors residents using privacy-preserving hardware — no cameras, no cloud. Falls among seniors (65+) are a leading cause of injury hospitalizations in Canada, rising **47% between 2008–2019**. Nursie addresses the gap between unwitnessed falls and delayed staff response.

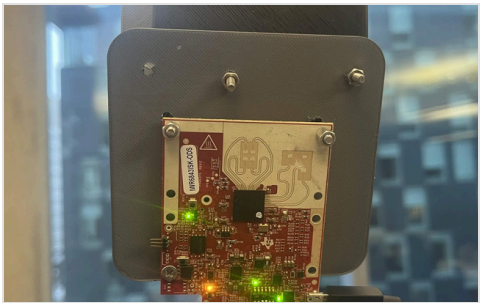
47%	3×	0	LAN
RISE IN FALL HOSPITALIZATIONS	SENSOR REDUNDACY	CAMERAS USED	LOCAL-ONLY ARCHITECTURE

PRIMARY STACK

`mmWave Radar` `ESP32 / FreeRTOS` `ROS 2` `MongoDB` `Gemini AI` `ElevenLabs` `OpenWRT`

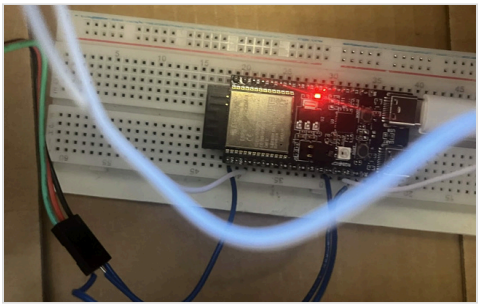
SENSOR HARDWARE

Privacy-Preserving · Redundant · Edge-Processed



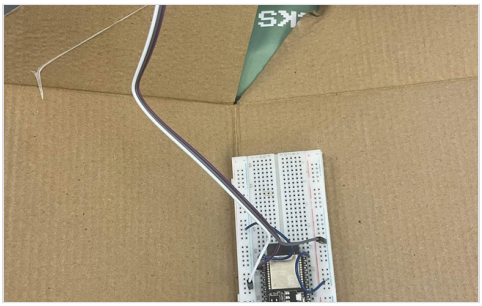
MMWAVE RADAR (IWR6843)

Continuous position & activity sensing via electromagnetic waves. Detects falls and tracks presence with no camera. Signal-processing tuned to reject curtain & object motion noise.



ESP32 DOOR + PIR NODE

Magnetic door sensor + PIR presence detector running FreeRTOS on ESP32. Publishes occupancy state over ROS 2 — flags night-time room exits for anomaly detection.



FLOOR VIBRATION MAT NODE

Detects floor-impact signature on contact. Fused with radar to distinguish a critical fall from a dropped object — reduces false-positive alerts for nursing staff.

DESIGN PRINCIPLES

Architecture Rationale

01 — FAST DETECTION FIRST

High confidence + fast alerting is the system's core mission. Every design choice subordinates itself to detection latency and reliability.

02 — SENSOR REDUNDANCY

mmWave radar, floor vibration mat, and wearable fused together. No single sensor failure breaks detection. Confidence scoring gates alert escalation.

03 — PRIVACY BY ARCHITECTURE

All data stays on a local OpenWRT LAN. No camera. No raw data leaves the facility by default. Compliant with Québec Law 25 privacy framework.

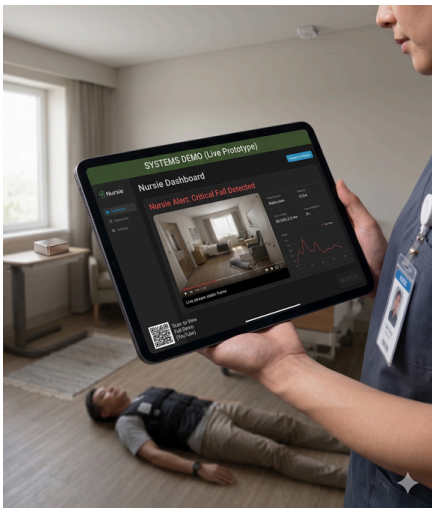
AI PIPELINE · DASHBOARD · OUTCOMES

ALERT PIPELINE & OUTCOMES

AI & BACKEND PIPELINE

Gemini · ElevenLabs · MongoDB · ROS 2

SENSORS	→	EDGE NODE	→	AI TRIAGE	→	PERSISTENCE	→	ESCALATION
mmWave Radar Vibration Mat Door / PIR Smartwatch		ESP32 FreeRTOS ROS 2 Messages OpenWRT LAN Signal Fusion		Gemini AI Intent / Priority ElevenLabs Voice Webhook Events		MongoDB Profile Anomaly Flags Behavior History Triage Logs		Phone Call Alert Dashboard Push Critical Priority Nurse Response



Watch Full Demo (Youtube Video)

Project Key Outcomes (Impact & Compliance)

- **Law 25 & Privacy-First:** Satisfies Québec's personal information protection framework (on-premises, zero-cloud dependency).
- **Advanced SNR & Fall Detection:** Fuses mmWave radar + vibration mat for soft vs. critical fall discrimination. Optimized Signal-to-Noise Ratio (SNR).
- **Production-Ready Edge Design:** Scalable, low-latency design on OpenWRT LAN. LAN-local by design.

The Engineering Story: Challenges & Solutions

SIGNAL CLARITY (mmWave SNR Optimization)

Differentiating human heartbeat signatures from moving curtains. Custom signal processing algorithms and threshold tuning.

Impact: Repeatable sensor node performance and manufacturable design.

REAL-TIME RELIABILITY & LAN MESH

Resolving real-time telemetry instability and sensor fusion logic through an optimized ESP32 mesh network.

Impact: Integrated sensor node ready for deployment in high-reliability environments.

Future Roadmap

Apple Watch Integration

Secondary vital-sign validation for high-risk patients.

Predictive Gait Analysis

Using radar data models to estimate fall risk before an incident occurs.

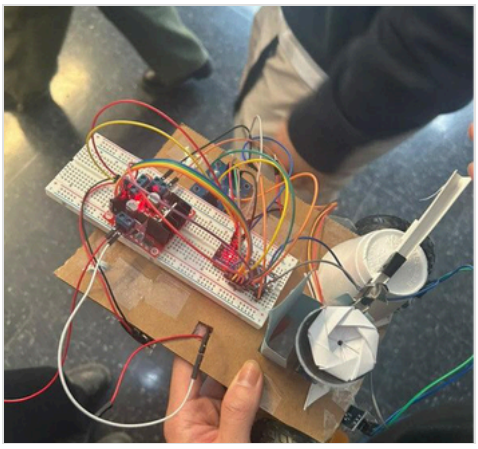
Medical-Grade Hardware

CHSLD rollout plan, with a focus on manufacturing efficiency and scalability.

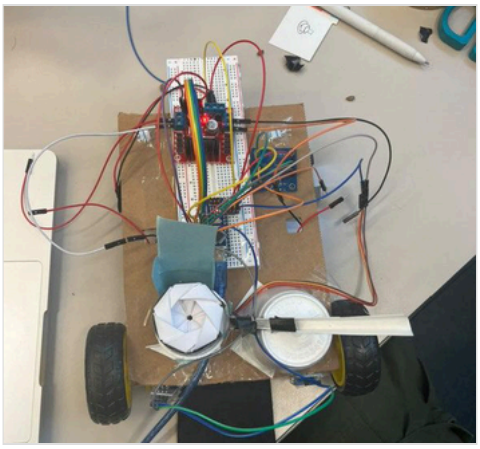
ROBOHACKS · FEB 2025 · ARDUINO · RECYCLED MATERIALS

🏆 WINNER — UNEXPECTED EXPEDITION AWARD · **5th Place Overall**

AUTONOMOUS REFORESTATION ROBOT



Top view — breadboard, Arduino, IR sensors & seed dispenser



Presentation demo — full assembly on competition floor

Built a fully autonomous seed-planting robot in **24 hours** using recycled materials—cardboard chassis, plastic bottles, and a paper-engineered seed dispenser. Coded the entire navigation and dispensing sequence in **C++ on Arduino**, wiring all electronics from scratch.

24h	3	5th
BUILD WINDOW	SENSORS INTEGRATED	PLACE OVERALL

STACK

Arduino C++ · IR Sensors · Color Sensor · DC Motors · Recycled Materials

ENGINEERING & CODE

Arduino · Sensors · Navigation

- **Autonomous navigation** — two infrared sensors fused with a color sensor for precise line-following along a designated black track
- **Motorized seed dispenser** — custom-engineered from paper and recycled materials; servo-actuated for accurate, timed seed placement
- **Full C++ codebase** — sensor polling loop, motor PID steering, and dispenser trigger sequence all written and debugged within the 24-hour window
- **Wire integration** — manually routed all electronics on a breadboard; confirmed signal integrity under live competition conditions

SUSTAINABLE DESIGN

Recycled-First Construction

RECYCLED-FIRST PRINCIPLE

The robot's chassis was built entirely from **repurposed cardboard and plastic bottles**. The seed dispenser was engineered from paper — folded into a functional rotating mechanism with no 3D-printed or machined parts. Demonstrated that **functional robotics hardware** can be prototyped with zero-waste materials under competition constraints.

✓ Designed, built & coded full robot independently in 24h

✓ Award-winning recycled-material robot construction

KEY COMPETENCIES DEMONSTRATED

Summary

DISCIPLINE	SCOPE	OUTCOME
Embedded Systems	Arduino C++, 3 sensors, motor control	Fully autonomous navigation & dispensing
Hardware Integration	Breadboard wiring, IR + color sensor fusion	Reliable line-following under competition load
Sustainable Fabrication	Cardboard, plastic bottles, paper dispenser	Zero-waste build — 🏆 Unexpected Expedition Award